DEFENSE LOGISTICS AGENCY
LAND AND MARITIME
P.O. BOX 3990
COLUMBUS, OHIO 43218-3990

DLA LAND AND MARITIME-VAI 18 August 2025

MEMORANDUM FOR MILITARY/INDUSTRY DISTRIBUTION

SUBJECT: Initial Drafts of various MIL-DTL-38999 Slash Sheets
(see table below for details).

These initial drafts for these subject documents are now available for viewing and downloading from the DLA Land and Maritime-VA Web site:

https://landandmaritimeapps.dla.mil/Programs/MilSpec/DocSearch.aspx

Initial Draft	Title	Project Number
MIL-DTL-38999/10C	Connectors, Electrical, Circular, Receptacle, Dummy Stowage, Bayonet Coupling, (MIL-DTL-27599, Series II and MIL-DTL-38999, Series II)	5935-2025-086
MIL-DTL-38999/20J	Connectors, Electrical, Circular, Receptacle, Threaded, Wall Mounting Flange, Removable Crimp Contacts, Series III, Metric	5935-2025-087
MIL-DTL-38999/21D	Connectors, Electrical, Circular, Threaded, Receptacle, Box Mounting Flange, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-088
MIL-DTL-38999/22K	Connectors, Electrical, Circular, Receptacle, Threaded, Dummy Stowage, Series III, Metric	5935-2025-089
MIL-DTL-38999/23D	Connectors, Electrical, Circular, Threaded, Receptacle, Jam-Nut Mounting, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-092
MIL-DTL-38999/24K	Connectors, Electrical, Circular, Receptacle, Threaded, Jam-Nut Mounting, Removable Crimp Contacts, Series III, Metric	5935-2025-093
MIL-DTL-38999/25D	Connector, Electrical, Circular, Threaded, Receptacle, Solder Mounting, Hermetic, Solder Contacts, Series III, Metric	5935-2025-094
MIL-DTL-38999/26H	Connectors, Electrical, Plug, Circular, Threaded, Straight, Removable Crimp Contacts, Series III, Metric	5935-2025-095
MIL-DTL-38999/27D	Connectors, Electrical, Circular, Threaded, Receptacle, Weld Mounting, Hermetic, Hermetic Solder Contacts, Series III, Metric	5935-2025-096
MIL-DTL-38999/28J	Connectors, Electrical, Circular, Nut, Hexagon, Connector Mounting, Series III and IV, Metric	5935-2025-097
MIL-DTL-38999/29D	Connectors, Electrical, Circular, Threaded, Plug, Lanyard Release, Fail-Safe, Removable Crimp Contacts, Pins, Series III, Metric	5935-2025-098
MIL-DTL-38999/30D	Connectors, Electrical, Circular, Threaded, Plug, Lanyard Release, Fail-Safe, Removable Crimp Contacts, Sockets, Series III, Metric	5935-2025-099

Major changes to these documents include removing dimensional equivalents.

Concurrence or comments are required at this Center within 30 days. Late comments will be held for the next coordination of the document. Comments from military departments must be identified as either "Essential" or "Suggested". Essential comments must be justified with supporting data. Military review activities should forward comments to their custodians of this office, as applicable, in sufficient time to allow for consolidating the department reply. Lack of response to this draft will be construed as concurrence.

If these documents are of interest to you, please provide your comments or suggested changes. The point of contact for this document is Mr. Bender, phone number 614-692-9185, facsimile transmission, 614-692-6939, e-mail Jacob.Bender@dla.mil or may be mailed via the US Postal Service to DLA LAND AND MARITIME, ATTN: VAI (Attention: Jacob Bender), P.O. Box 3990, Columbus, OH 43218-3990.

Sincerely,

/ SIGNED /

ERIC CANDELARIA
Acting Chief,
Interconnection Branch

CC: David Devine, Robert Schoening, Alan Drais, Eddie Shen, Deborah Thompson

Note: This initial draft, dated August 22, 2025, prepared by DLA-CC has not been approved and is subject to modification. DO NOT USE FOR ACQUISITION PURPOSES. (Project: 5935-2025-087)

METRIC

MIL-DTL-38999/20J
DRAFT
SUPERSEDING
MIL-DTL-38999/20H
w/ AMENDMENT 1
11 August 2022

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, RECEPTACLE, THREADED, WALL MOUNTING FLANGE, REMOVABLE CRIMP CONTACTS, SERIES III, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

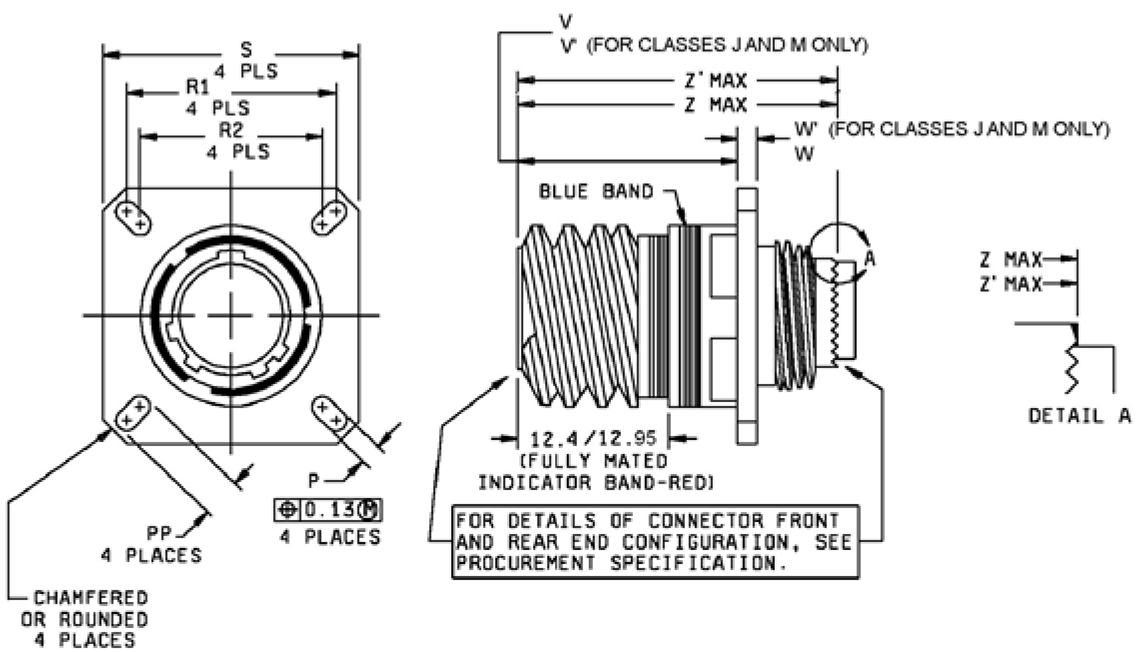

FIGURE 1. Receptacle.

AMSC NA FSC 5935

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Shell size	Shell size code	M max, (see note 5)	P ± 0.20	PP ± 0.20	R1 (see note 3)	R2 (see note 4)	S ± 0.3
9	A	5.80	3.25	5.49	18.26	15.09	23.8
11	B		3.25	4.93	20.62	18.26	26.2
13	C		3.25	4.93	23.01	20.62	28.6
15	D		3.25	4.39	24.61	23.01	31.0
17	E		3.25	4.93	26.97	24.61	33.3
19	F		3.25	4.93	29.36	26.97	36.5
21	G	5.00	3.25	4.93	31.75	29.36	39.7
23	H		3.91	6.15	34.93	31.75	42.9
25	J		3.91	6.15	38.10	34.93	46.0

Shell size	Shell size code	V + 0.00 - 1.25	V' + 1.4 - 0.0	W	W'	Z max	Z' max	External bending moment, minimum, classes J and M
9	A	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	8.865
11	B	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	29.492
13	C	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	34.485
15	D	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	40.566
17	E	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	50.781
19	F	20.83	19.5	2.5/2.1	3.65/2.1	31.5	32.0	51.450
21	G	20.07	18.7	3.2/2.1	4.35/2.1	31.5	32.0	80.941
23	H	20.07	18.7	3.2/2.1	4.35/2.1	31.5	32.0	69.810
25	J	20.07	18.7	3.2/2.1	4.35/2.1	31.5	32.0	91.310

NOTES:

1. Dimensions are in millimeters.
2. Front or rear panel mounting.
3. Front panel mounting only.
4. Front or rear panel thickness. To ensure full mating of the plug or the backshell, it is recommended that the front or rear mounting panel thickness not exceed the M values specified in the table.
5. The specified red band location is measured from the front edge of the shell to the back edge of the red band.
6. Dimensions V', W', and Z' are for classes J and M, only. Dimensions V, W, and Z are for all other classes.

FIGURE 1. Receptacle – Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1. Interface dimensions shall be in accordance with MIL-DTL-38999.

This connector mates with MIL-DTL-38999/26.

For insert arrangements: See MIL-STD-1560.

Connector accessories: See SAE-AS85049.

Part or Identifying Number example:

<u>D38999/</u>	<u>20</u>	<u>W</u>	<u>A</u>	<u>35</u>	<u>P</u>	<u>N</u>

<u>D38999/</u>	DoD number prefix
<u>20</u>	Specification sheet number
<u>W</u>	Class
<u>A</u>	Shell size code
<u>35</u>	Insert arrangement
<u>P</u>	Contact style
<u>N</u>	Polarizing positions

Restrictions: Class F is not for Navy use and is inactive for Air Force new design use.

NOTE: The term PIN is equivalent to the term which was previously used in this specification.

Qualification required: The activity responsible for the qualified product list for this specification sheet is DLA Land and Maritime-VQ, 3990 East Broad Street, Columbus, Ohio, 43218-3990.

Note: Bending moment torque . Bending moment requirements were determined through finite element analysis and testing parts to yield strength. Material yield strength was characterized by a normal distribution with $\mu_{material,yield} = 275.79 MPa$ and $\sigma_{material,yield} = 13.78\ MPa$. A yield stress divided by yield torque ratio and a real world torque variation were determined and given in Table I. Requirements were set at 3 standard deviations from the mean as given below:

$$\tau_{requirement} = \frac{\mu_{material,yield}}{\alpha_{yield}} - 3 * \sqrt{\frac{{\sigma_{material,yield}}^2}{{\alpha_{yield}}^2} + {\sigma_r}^2}$$

TABLE I. Bending moment torque parameters

Shell Size	$\alpha_{yield}\left(\frac{MPa}{N*m}\right)$	$\sigma_r(N*m)$
9	25.362	0.391
11	7.540	1.494
13	6.610	1.213
15	4.620	5.636
17	4.386	2.527
19	4.165	3.642
21	2.686	5.116
23	2.669	9.902
25	2.270	8.022

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-STD-1560
MIL-DTL-38999/26
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy - AS
Air Force – 85
DLA - CC

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19

Preparing activity:
DLA - CC

(Project: 5935-2025-087)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.